

Arduino Quiz

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Quiz Instructions

A few questions to test your understanding of the Arduino and our electronics interfaces to it. Unless stated otherwise, assume the Arduino's analog reference pin, $V_{ref} = 4.50$ VDC.

Question 1

$V_{ref} = 4.5V$ and 3.0VDC is on Arduino pin A0. `analogRead(A0)` will return what value?

Question 2

The LM317 is adjusted to 10VDC output, and connected directly to Arduino pin A0. `analogRead(A0)` returns

- A. 852
- B. 1023
- C. 0
- D. Probably nothing because the Arduino is fried, caput, dead ...

Question 3

The LM317 is adjusted to 8VDC output, and connected to a voltage divider consisting of $R1 = 2k$ and $R2 = 1k$ to ground. What is the voltage across $R2$ (in volts)?

Question 4

The LM317 is adjusted to 8VDC output, and connected to a voltage divider consisting of $R1 = 2k$ and $R2 = 1k$ to ground. A0 is connected to $R2$, `analogRead(A0)` returns what value (in ADU)?

Question 5

The LM317 is adjusted to 8VDC output, and connected to a voltage divider consisting of $R1 = 2k$ and $R2 = 1k$ to ground. A 1kohm load is connected in parallel with the voltage divider and the LM317 (draw a schematic). What is the voltage across $R2$ (in volts)?

Question 6

The LM317 is adjusted to 8VDC output, and connected to a voltage divider consisting of $R1 = 2k$ and $R2 = 1k$ to ground. A 1kohm load is connected in parallel with $R2$ of the voltage divider (draw a schematic). What is the voltage across $R2$ (in volts)?

Question 7

The LM317 is connected to a voltage divider consisting of $R1 = 2k$ and $R2 = 1k$ to ground, and A0 = the voltage across $R2$. If `analogRead(A0)` returns 345, what is the LM317 output voltage (in volts)?

Solutions

1. $3V * (1023\text{ADU}/4.5V) = 682\text{ADU}$
2. **D**
3. $V_2 = 8V * (1k/1k+2k) = 8/3V = 2.68V$
4. $2.68V(*1023\text{ADU}/4.5V) = 606\text{ADU}$
5. $2.68V$
6. $8V * .5k/(2k+.5k) = 1.6V$
7. $345\text{ADU}*(4.5V/1023\text{ADU})*(2k+1k)/1k = 4.55V$